

Raisul Kabir News

Lecturer, Computer Science and Engineering, BRAC University

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Education

BRAC University

M.Sc. in Computer Science and Engineering (CGPA: 3.85/4.00)

Ongoing

- To Be Completed: 2026

B.Sc. in Computer Science and Engineering (CGPA: 3.91/4.00)

2019 – 2022

- Completed: 2022

Publications

An Advanced Data Fabric Architecture Leveraging Homomorphic Encryption and Federated Learning 2024

Published in Information Fusion (Elsevier, IF: 17.4)

- Secure medical image analysis using federated learning and homomorphic encryption.
- Enables privacy-preserving collaborative deep learning in healthcare.

Interpretable Deep Learning Approaches for Reliable GI Image Classification: A Study with the HyperKvasir Dataset 2024

Published in 27th International Conference on Computer and Information Technology (ICCIT)

- Developed interpretable deep learning models for medical image classification.

Ongoing Researches

Why Cross-Dataset Transfer Fails in Human Activity Recognition (HAR): A Confound-Controlled Analysis of Classifier Degeneration 2026

Submitted in ACM(IMWUT)

- Confound-controlled evaluation protocol identifying four methodological errors in prior HAR cross-dataset transfer studies, including a previously-unreported PAMAP2 sensor-channel loading error.
- Introduces a label-free prediction-entropy diagnostic and per-class CORAL alignment baseline for deployable wearable HAR systems.

Verifiable Federated Aggregation as a Native Data Fabric Primitive (zk-FedMax) 2026

Ongoing

- Extending prior data fabric architecture by introducing zero-knowledge proofs for selection-based federated aggregation (FedMax/FedMin/top-k), addressing the untrusted-server assumption left open in earlier work.
- Formalizing ZK-provable comparison/argmax operations for aggregator honesty, without revealing individual client updates.
- Targeting convergence guarantees under non-IID settings and empirical validation across multiple healthcare imaging datasets.

Quantifying Urban Heat Island Dynamics Using Spatio-Temporal Machine Learning Models 2026

Ongoing under my supervision

- Spatio-temporal forecasting of surface urban heat island intensity in Dhaka using multi-sensor Landsat/MODIS satellite data and ConvLSTM deep learning.
- Enables data-driven identification of heat-vulnerable urban zones and evaluation of vegetation-based mitigation strategies for rapidly urbanizing megacities.

Research Skills

Federated Learning and Distributed Learning Systems

- Designed and implemented decentralized learning frameworks enabling collaborative model training without centralized data sharing.
- Applied in privacy-preserving medical image analysis as part of a Q1 journal publication in *Information Fusion*.

Privacy-Preserving Machine Learning

- Applied partially homomorphic encryption techniques to secure sensitive data during distributed model training and inference.
- Addressed data locality, security, and regulatory constraints in healthcare-oriented AI systems.

Interpretable and Explainable Artificial Intelligence (XAI)

- Integrated explainability techniques to improve transparency and clinical trust in deep learning models.
- Applied interpretability methods in GI image classification using the HyperKvasir dataset (IEEE conference publication).

Automated Testing and Program Analysis

- Applied property-based testing, fuzzing, and automated program analysis to systematically explore software behavior and identify reproducible defects.

Experimental Design and Benchmarking

- Designed controlled experiments for both centralized and federated learning settings.
- Conducted systematic benchmarking and comparative analysis for peer-reviewed research reporting.

Secure and Scalable AI System Design

- Designed secure and automated AI-enabled systems with an emphasis on scalability and access control.
- Demonstrated through development of a secure Discord-based course management platform for academic coordination.

AI for Real-World, Ethical, and Societal Impact

- Developed an understanding of applying AI techniques to real-world problems with societal impact, including healthcare analytics and urban heat risk analysis.
- Gained knowledge of ethical and responsible AI practices—such as data privacy, transparency, and fairness through relevant coursework and professional workshops.

Professional Experience

BRAC University, Department of Computer Science and Engineering **May 2024 – Present**
Lecturer and Program Coordinator (Microprocessor Course) Dhaka, Bangladesh

- Conducted undergraduate courses on programming languages (Python, Java), microprocessors, Algorithms, and data structures. Coordinated theory instruction, designed course contents and examinations.

Daffodil International University, Department of ITM **Jul 2023 – May 2024**
Lecturer Dhaka, Bangladesh

- Conducted undergraduate courses on Human-Computer Interaction and Artificial Intelligence. Advised and mentored students, and evaluated lab performances, assignments, and examinations.

BRAC University, Department of Computer Science and Engineering **Sept 2023 – Apr 2024**
Adjunct Lecturer Dhaka, Bangladesh

- Conducted laboratory courses including Algorithms Lab (CSE 221), Microprocessors Lab (CSE 341), and Programming Language I Lab (CSE 110).

BRAC University, Department of Computer Science and Engineering **Jul 2021 – Dec 2022**
Student Tutor / Teaching Assistant Dhaka, Bangladesh

- Assisted in core undergraduate courses including Microprocessors (CSE 341), Digital Logic Design (CSE 260), Algorithms (CSE 221), and Programming Language I (CSE 110).

TechShoi **Apr 2022 – Jul 2022**
Junior Software Engineer Dhaka, Bangladesh

- Developed backend architecture using web frameworks and RESTful APIs for a large-scale enterprise system in Bangladesh.

Achievements

Quality Journal Publication Award

- Awarded in 2025 by BRAC University for publishing a research article in a top 1% Q1 journal.
- Journal: *Information Fusion*, Elsevier (Impact Factor: 17.4).

Open-Source Contribution

- Identified and reported a reproducible numeric-literal parsing bug in `tomlc17`; the issue was confirmed and fixed upstream (commit `e229ff5`).

Vice Chancellor's and Dean's List Honoree

- Awarded for academic excellence during undergraduate studies.
- Vice Chancellor's List: Fall 2019, Spring 2020, Fall 2020, Spring 2021, Summer 2021, Spring 2022.
- Dean's List: Summer 2019, Summer 2020.

Cultural Representation at BRAC University

- Awarded for outstanding performance in a cultural drama at the BRAC University.(Summer 2019)

Google Code Jam Participant

- Participated in international competitive programming contests organized by Google.(Years: 2021 and 2022.)

Assembly Language Programming Milestone

- Solved over 300 Assembly language programming problems.
- Demonstrated strong proficiency in low-level programming and computer architecture (2020–2022).

Projects

Agentic Property-Based Fuzzing of tom17 | LLM Agents, Property-Based Testing, Program Analysis

- Developed an agentic property-based testing framework that iteratively generates and refines Hypothesis strategies for testing the tom17 TOML parser.
- Integrated grammar-aware generation, sanitizer-based C harnesses, oracle-guided feedback, and automated failure triage to explore parser behaviors.
- Discovered and reproduced a numeric-literal conformance bug that was subsequently confirmed and fixed in the upstream tom17 repository.

Urban Heat Mapping and Prediction System | AI, Machine Learning, Geospatial Analysis

- Developed an AI-based classification and prediction framework to analyze urban heat distribution using environmental and spatial datasets.
- Applied machine learning models to classify high-risk urban heat zones and predict heat intensity patterns.
- Supported climate-aware urban planning through data-driven heat risk analysis and visualization.

Secure Discord-Based Course Management Bot | Automation, Discord API, Backend Systems

- Designed and developed a secure, automated Discord bot to manage course communication and coordination as a Program Coordinator.
- Implemented role-based access control, automated announcements, and structured discussion channels to ensure secure and organized student communication.
- Enhanced course management efficiency by centralizing academic updates, student interactions, and administrative workflows on a single platform.

Skills

Soft Skills

- **Independent Research Discipline** — Developed the ability to plan, execute, and complete long-term research tasks independently through peer-reviewed journal and conference publications.
- **Scientific Communication and Argumentation** — Practiced articulating complex technical ideas clearly through academic writing, classroom teaching, and research presentations to diverse audiences.
- **Leadership and Research Coordination** — Demonstrated leadership as a Program Coordinator and football team coach by organizing teams, managing responsibilities, and guiding individuals toward collective outcomes.
- **Collaboration in Academic Environments** — Worked collaboratively with faculty members, co-authors, teaching assistants, and students in multi-role academic settings.
- **Resilience and Time Management** — Balanced research, teaching, coordination duties, and professional development under academic deadlines and performance expectations.

Hard Skills

- **Programming and Scientific Computing** — Python, Java, C++, C#, Dart, Assembly, Numerical computation and data processing.
- **AI Frameworks and Research Tools** — PyTorch, NumPy, Pandas; model training, evaluation, and experimentation pipelines.
- **Web, Backend, and Automation Systems** — Django, REST APIs, HTML, CSS; development of secure and automated academic platforms.

- **Databases and Development Toolchain** — MySQL, MongoDB, Git, Linux, LaTeX, Android Studio.

Languages

- **English:** Full Professional Proficiency
IELTS Academic: Overall Band 7 (Listening: 7.5, Reading: 7.5, Writing: 6.5, Speaking: 6.5)
- **Bangla:** Native proficiency
- **Hindi:** Limited working proficiency

Professional Training

Outcome-Based Education (OBE) and Bloom's Taxonomy

- Focus on CLO–PLO mapping, assessment design, and curriculum alignment in higher education.
- Instructor: Prof. Dr. Md. Mozahar Ali (Bangladesh Agricultural University);

Online Teaching Tools and Digital Pedagogy

- Training on LMS usage, virtual assessments, and student engagement in online and hybrid classrooms.
- Instructor: Mr. Narayan Ranjan Chakraborty, Associate Professor and Associate Head, CSE (Daffodil International University);

AI Education and Applications

- Covered AI fundamentals, applied machine learning use cases, and ethical considerations in education.
- Instructor: Mr. Amit Chakraborty, Assistant Professor, CSE (Daffodil International University)

Leadership and Professional Skill Development

- Emphasis on academic leadership, communication skills, teamwork, and professional growth.
- Instructor: Dr. Md. Sabur Khan, Chairman, Board of Trustees (Daffodil International University);

Extracurricular

Soccer Team of Information Technology and Management | Coach

ITM Department of Daffodil International University

Sept 2023 – Apr 2024

- Conducted training on possession play, pressing tactics, and counter-attacking strategies; achieved runners-up positions in two inter-university tournaments.

CSE Department of BRAC University

Jul 2026 – Jul 2026

- Participated in small inter-university tournament.

BRAC University Computer Club and Robotics Club (BUCC) and (ROBU) | Member

References

Dr. Md. Golam Rabiul Alam

Professor, and Thesis Coordinator, Dept. of Computer Science and Engineering, BRAC University

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